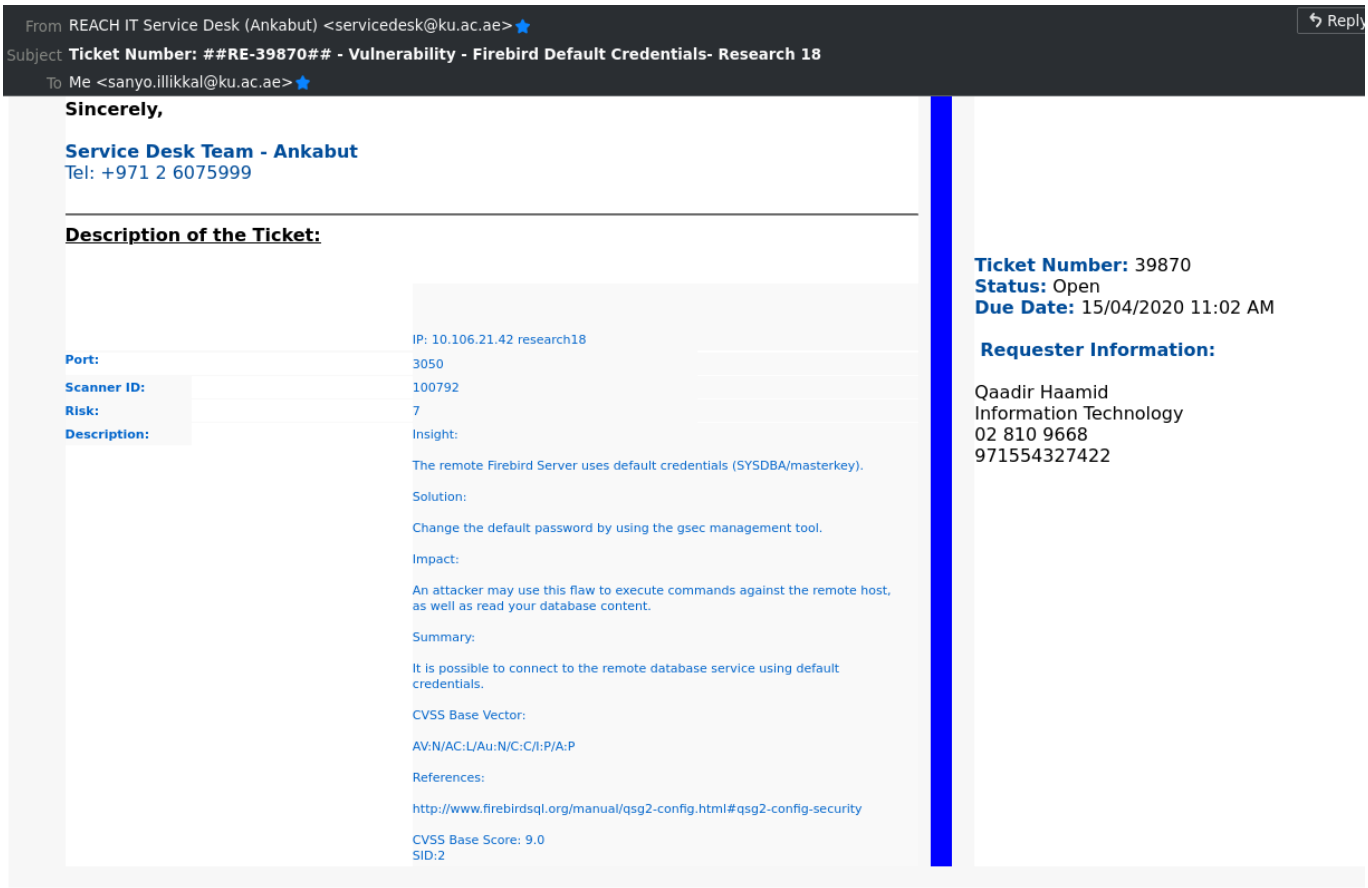
Patches update policy & procedure.

Research Servers

All the Research Servers are installed with specific version of OS packages supported by the End User Application. Regular system update will cause OS package upgrade and can break the Application dependence or end user workflow.
We will only perform the OS Patches update for specific Vulnerability reported by IT Security Team.

IT Security team will be regularly performing System Log Monitoring and Vulnerability scan on Research Systems. Tickets will open for Vulnerability reported. Necessary actions will be taken for the same by the Team.

Sample Ticket Screenshot



From REACH IT Service Desk (Ankabut) <servicedesk@ku.ac.ae>
Subject **Ticket Number: ##RE-39870## - Vulnerability - Firebird Default Credentials- Research 18**
To Me <sanyo.illikkal@ku.ac.ae>

Sincerely,

Service Desk Team - Ankabut
Tel: +971 2 6075999

Description of the Ticket:

IP: 10.106.21.42 research18

Port:	3050
Scanner ID:	100792
Risk:	7
Description:	Insight:

The remote Firebird Server uses default credentials (SYSDBA/masterkey).

Solution:

Change the default password by using the gsec management tool.

Impact:

An attacker may use this flaw to execute commands against the remote host, as well as read your database content.

Summary:

It is possible to connect to the remote database service using default credentials.

CVSS Base Vector:

AV:N/AC:L/Au:N/C:C/I:P/A:P

References:

http://www.firebirdsql.org/manual/qsg2-config.html#qsg2-config-security

CVSS Base Score: 9.0
SID:2

Ticket Number: 39870
Status: Open
Due Date: 15/04/2020 11:02 AM

Requester Information:

Qaadir Haamid
Information Technology
02 810 9668
971554327422

From IT Service Desk <servicedesk@ku.ac.ae>
Subject **Ticket Number: ##RE-89025## - FW: FortiEDR Security Incident 5273512 Notification - EventsQueue | Classification: PUP | Device: KU2RPAAWV00**
To Me <sanyo.illikkal@ku.ac.ae>

From: Llallum Victoria
Sent: Friday, July 9, 2021 3:42:50 PM
To: mustafa.jama@ku.ac.ae; qaadir.haamid@ku.ac.ae
Cc: FortiResponder-Team
Subject: FortiEDR Security Incident 5273512 Notification - EventsQueue | Classification: PUP | Device: KU2RPAAWV007

Hi Khalifa University,

FortiEDR has detected the execution of process DWRCS.exe on one of your devices. DameWare is a remote desktop administration tool that is not inherently malicious but can be leveraged by attackers to move across networks and access internal resources. If this software is permitted in this environment please let us know and we will create an appropriate exception.

Event ID: 4518186
Command Line: -service
Hostname: KU2RPAAWV007
Event Type: TCP Listen
File Path: \Device\HarddiskVolume2\Windows\SysWOW64\DWRCS.EXE

Command Line: -service
Hostname: KU2RPAAWV007
Event Type: Service Access
File Path: \Device\HarddiskVolume2\Windows\SysWOW64\DWRCS.EXE

Command Line: -service
Hostname: KU2RPAAWV007
Event Type: Connection
Destinations: 127.0.0.1
File Path: \Device\HarddiskVolume2\Windows\SysWOW64\DWRCS.EXE

Recommendations:
Removal is recommended if no legitimate business use can be validated.

Regards,
lvictoria

From REACH IT Service Desk (Ankabut) <servicedesk@ku.ac.ae>
Subject **Ticket Number: ##RE-44226## - MI-GEODB- 10.106.0.211 - Postgresql default password**
To Me <sanyo.illikkal@ku.ac.ae>
Reply Forw

To protect your privacy, Thunderbird has blocked remote content in this message.

Description of the Ticket:

Request the following task to be completed by the Research Admins, Network Admin and KU DBAs:

1. default postgres password to be changed.- DBA
2. database baselined remove all default accounts- DBA
3. set host based firewall to only allow access to db on port 5432 from allowed web or application servers- Research Admin
4. set firewall and NAC policy to only allow access to server from valid management ranges(ssh/22) and the db on port 5432 from allowed web or application servers only - Network

10.106.0.211 - mi-geodb-prod

Reported Ports
22/tcp 5432/tcp

Vuln Name	VulnID	Service	Severity
PostgreSQL weak password	103552	postgres (5432/tcp)	Critical

Vulnerability Detection Result:

It was possible to login as user postgres with password "postgres".

Solution:

Change the password as soon as possible.

Summary:

It was possible to login into the remote PostgreSQL as user postgres using weak credentials.

CVSS Base Vector:

Family name: Default Accounts

Category: attack

Copyright: This script is Copyright (C) 2012 Greenbone Networks GmbH

Summary: NOSUMMARY

Version: 2020-01-28T13:26:39+0000

Ticket Number: 44226
Status: Open
Due Date: Jun 7, 2020 12:56 PM

Requester Information:

Qaadir Haamid
Information Security
02 810 9668
971554327422

All the Research Server OS will be upgraded to the latest OS based on the OS end of life and latest Application requirements.

HPC Systems

HPC Server OS Patches will be released by HPC Support vendor and Research Team will review and plan for the upgrade.

Recent OS upgrade Screenshot

From Me <sanyo.illikkal@ku.ac.ae>
Subject **Re: Upgrade to CentOS 8.4 and latest OpenHPC 2.3 repo**
To rf@q-leap.de
Cc Dureid El Moghraby <dureid.elmoghraby@ku.ac.ae>, Mohamed Mekias <mmekias@hpcme.com>, Sergi

Dear Roland,

We are fine with the OS upgrade.

Kindly plan and let us know the date scheduled for the activity.

Below is our action plan:

1. OS migration :
   - Master node and Storage nodes - Qlustar team
2. Reboot nodes - by KU Team.
   - Login node (login-3)
   - Compute Nodes (cn-09-31 to cn-09-34)
   - GPU nodes (gpu-11-3)
3. Test the applications in new OS - by KU Team.
4. Reboot all compute nodes as part of rollout new OS. - by KU Team.